

# Karuna: A Community Driven Smartphone Application for Location Aware Image Based Injury Detection, Reporting and Rescue of Injured Animals

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## Abstract

Urban animal rescue is constrained by poor coordination rather than limited resources. Injured animals are reported through fragmented channels, rescue response times stretch to 24 hours or more, and no single system connects reporting, treatment, donations, and outcomes for an individual case. This work proposes KARUNA, a unified, AI-assisted smartphone platform to address these gaps. When a citizen photographs an injured animal, the application (app) automatically captures GPS and timestamp data, then passes the image to a YOLOv8-based detection model fine-tuned on Indian street animal injury types. A hybrid triage engine combining veterinary decision rules with gradient-boosted classification produces a severity score and multilingual first-aid guidance. That severity score feeds directly into an automated matching algorithm that dispatches the nearest qualified responder without manual intervention. All subsequent events, including collection, treatment, discharge, donation, and adoption, are logged against a single case record and shared with all stakeholders in real-time. Per-case donation tracking follows the transparency principles used by blockchain-based charitable systems, while avoiding blockchain infrastructure complexity through an append-only per-case event log. The platform is designed for multi-organisation, multi-city deployment, generating standardised, georeferenced incident data that can directly support municipal ABC programme planning and enable citizen-driven animal health surveillance.

## 1 Problem Statement

Walk through any Indian city on a Sunday morning and the odds are good that you will pass a dog or a cow lying injured on the roadside. What happens next depends entirely on luck. If someone stops, they will likely call a number they found online, wait, call another, and eventually give up or take matters into their own hands. There is no single number to call. There is no guarantee anyone will come. The animal may wait hours, or it may not survive the wait at all.

This is not a resource problem. NGOs exist. Veterinary clinics exist. Volunteers exist in most cities. The problem is that none of these actors can see each other. A rescue NGO in Vijayawada has no real-time view of which cases its volunteers are already handling. A donor who sent money last month has no idea whether it reached the animal they saw in a Facebook post. A municipal ABC officer planning sterilisation routes has no incident data to work from. Everyone

is trying, but everyone is working blind. The result is a system that is simultaneously overstretched and underused. The same case gets reported three times while another one goes unnoticed entirely, and the animals pay for the gap with their lives.

## 2 Literature Review and Gaps

The research landscape here splits into three reasonably distinct threads, though none of them quite connects to the others. The first component is citizen-facing reporting. In [1], researchers showed that a structured web portal could replace informal phone calls and route incidents to the right authority. This is a genuine step forward; however, the system is limited to a web-based interface, lacks GPS and image capture capabilities, and stops at the routing stage. In [2], researchers embedded an AI chatbot into a mobile reporting app that demonstrably reduce triage latency and lowered the load on human dispatchers. In the deployed space, platforms like Let It Wag [3], Strayz [4], and PetFinder [5] handle basic reporting and NGO routing, but none of them touch treatment tracking, per-case donations, or anything downstream of the initial handoff. In the Indian context, Researchers in [6] developed HSIApps, a mobile and web dashboard for street dog sterilisation programmes that digitises field data capture, and in [7] researchers separately modelled free-roaming dog population management through systems dynamics; both contributions advance specific operational segments but neither offers an integrated, citizen-facing rescue-to-outcome platform.

The second thread is computer vision for animals. In [8], researchers benchmarked a cascaded YOLOv8 system on animal detection and reported 97% accuracy across species using adaptive preprocessing and multi-feature extraction. In [9], researchers developed a YOLOplusUNet deep learning pipeline for automated wound detection and size estimation in animal subjects, demonstrating reliable localisation of wound boundaries. None of these, however, were designed for the urban stray context in India, and none feed their outputs into any kind of triage or dispatch pipeline. On the triage methodology side, in [10], researchers Fernandes et al. demonstrated that supervised ML models can match trained nurses on patient triage severity classification. These are human medicine papers, but the hybrid ruleplusML approach they validate is directly applicable here.

The third thread is donation transparency and fund accountability. In [11], researchers built a sustainable blockchain framework for the financial support of charitable organisations and showed it meaningfully improved donor confidence. In [12], researchers went a step further by proposing a comprehensive blockchain-based charity collection framework with smart contracts, an economic model, and CharityCoin for end-to-end audit trail. In [11], researchers demonstrated transparent public fund tracking using a blockchain-based charity donation service system during COVID19. Separately, in [13], researchers used CatBoost models to predict shelter adoption timelines, which is a useful data point for how analytics can drive animal welfare decision making.

What none of this adds up to is a complete picture of a single case. In [14], researchers are direct about this: most digital systems for zoonotic disease surveillance are built for institutional or passive use, not for citizen-driven realtime incident management. In [15], researchers reach a similar conclusion from the animal health surveillance side; their scoping review finds real gaps in multisector, citizen-driven data collection that current systems simply do not address. The throughline is obvious: reporting is handled separately from treatment, treatment from donations, and donations from any kind of research output. No one has built the thing that connects all of them for a single animal, from the moment someone finds it on the street to whatever happens to it next.

The authors reviewed existing literature in this domain to analyse current approaches and identify the research gaps. Based on functionality, the works are grouped into five categories: (a) Citizen reporting and rescue coordination, (b) Field operations and programme management, (c) AI detection, triage, and prediction, (d) Financial transparency and donation systems, and (e) Population modelling and health surveillance.

The authors have reviewed the existing works on the systems dealing with citizen reporting and rescue in Table- 1. But there is no end-to-end rescue reporting (by citizens), rescue, triage, treatment, management system. The authors have also reviewed the existing works on Field operations and programme management, which is presented in Table- 2. but, the system did not consider citizen facing rescue reporting, AI injury detection triage, automated dispatch and donation tracking. In Table- 3, the authors have compared the existing works on AI prediction, detection, triage. However, the works are not applicable in real rescue system. In Table- 4, authors have shown different existing works on finance and donation systems. But these works are only financial. There is no relation with rescue cases or results. A comparative study on population modelling and health surveillance is given in Table- 5. In this area, available research primarily focus

on aggregate information. There is no real-time action or tracking. In short, there is research in this field. But, End-to-end integrated system is still needed. A cyber-physical rescue ecosystem is still needed in this area.

Table 1: Literature review on citizen reporting and rescue coordination

| Sl. No | Ref. | Method used                                      | Key Contribution                                                                             | limitations                                                                                             |
|--------|------|--------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1      | [1]  | Web-based reporting portal                       | Structured digital intake replaces adhoc phone reporting; centralised routing to authorities | Web only; no GPS or image capture; no automated severity scoring; no treatment tracking                 |
| 2      | [2]  | AI-assisted reporting Mobile app with AI chatbot | AI chatbot reduces triage latency and improves report quality                                | No treatment management, donation tracking, or adoption stage; chatbot only, no visual injury detection |
| 3      | [3]  | Mobile app with geotagging                       | Image-based incident reporting, NGO routing, and adoption listing                            | No AI triage; no treatment tracking; partnerNGO dependency limits coverage                              |

Table 2: Literature review on Field operations and programme management

| Sl. No | Ref. | Method used                    | Key Contribution                                                                                                                    | limitations                                                                                                                            |
|--------|------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1      | [6]  | HSIApps mobile + web dashboard | Android mobile app and web dashboard for field teams during street dog sterilisation; digital data capture in Indian ABC programmes | Sterilisation-focused; no citizen-facing rescue reporting; no AI injury detection, triage, or automated dispatch; no donation tracking |

Table 3: Literature review on AI detection, triage, and prediction

| Sl. No | Ref. | Method used     | Key Contribution                                                                                               | limitations                                                                                                      |
|--------|------|-----------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1      | [8]  | Cascaded YOLOv8 | 97% accuracy wildlife detection using cascaded YOLOv8 with adaptive pre-processing and feature extraction      | Applied to wildlife, not urban street animals; no injury classification; not integrated with any rescue workflow |
| 2      | [9]  | YOLO + UNet     | Automated wound detection and size estimation using deep learning; YOLObased localisation pipeline             | Laboratory mouse context; no urban stray injuries; no triage engine; not integrated with rescue or dispatch      |
| 3      | [10] | Supervised ML   | MLassisted triage achieves performance comparable to trained nurses; hybrid rule + ML outperforms either alone | Human medicine context; no animal welfare application; no imagebased or species specific triage                  |
| 4      | [13] | CatBoost ML     | Predicted shelter pet adoption timelines; ranked feature importance                                            | Focused on shelterhoused animals; no street animal context; no realtime rescue integration                       |

### 3 Objective

To develop a unified, AI assisted digital platform (KARUNA) that closes the current gaps in urban animal welfare by integrating image and GPS-based citizen reporting, automated AI triage and First aid, coordinated multistakeholder rescue and treatment workflows, and transparent, case wise donation and adoption tracking across multiple Indian cities.

## 4 Proposed Idea: KARUNA A Unified AI Assisted Rescue Platform

The core idea behind KARUNA is straightforward: one case record, one platform, all the way through. A citizen finds a dog bleeding on the street. That moment the photograph, the GPS pin, the timestamp is the start of a record that stays alive until the animal is either adopted, released, or lost. Every actor who touches that case sees the same record. Nothing gets lost in a forwarded WhatsApp message. Nothing starts over when the animal moves from the street to a clinic.

The overview of the proposed system is presented in Fig. 1. A member of the public reports an injured animal via a mobile phone with geolocation and date/time. An YOLOv8-based model automatically identifies injuries and severity.

Table 4: Literature review on Financial transparency and donation systems

| Sl. No | Ref. | Method used      | Key Contribution                                                                                         | Limitations                                                                         |
|--------|------|------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1      | [11] | Blockchain       | Sustainable blockchain framework for financial support of charitable organisations using smart contracts | Financial layer only; no rescue operations or case outcomes linked                  |
| 2      | [12] | Blockchain + ICO | End to end charity collection framework using blockchain with Charity-Coin and smart contracts for audit | Generic charity domain; no percase audit trail linked to individual rescue outcomes |

Table 5: Literature review on Population modelling and health surveillance

| Sl. No | Ref. | Method used                | Key Contribution                                                                                                                        | Limitations                                                                                                                        |
|--------|------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 1      | [7]  | Systems dynamics modelling | Systems model comparing dog population management strategies; validates combined fertility control and responsible ownership approaches | Population-level modelling only; no individual case tracking, citizen-facing rescue coordination, or real-time incident management |
| 2      | [14] | Zoonotic surveillance      | Modern technologies for zoonotic disease surveillance; GPS and app-based reporting discussed                                            | Review scope is institutional surveillance; no citizen facing rescue coordination or case management workflow                      |
| 2      | [15] | Animal health surveillance | Comprehensive review of syndromic animal health surveillance; identifies gaps in citizen-driven data collection                         | Focused on institutional surveillance systems; no real-time citizen reporting, rescue coordination, or donation tracking           |

The system provides a severity score and multilingual first-aid guidance. Responders are automatically dispatched to the location based on urgency. An unified dashboard is used to manage all case details such as, treatment updates, and donation. Fig. 2 shows the KARUNA Citizens/Reporters interface.

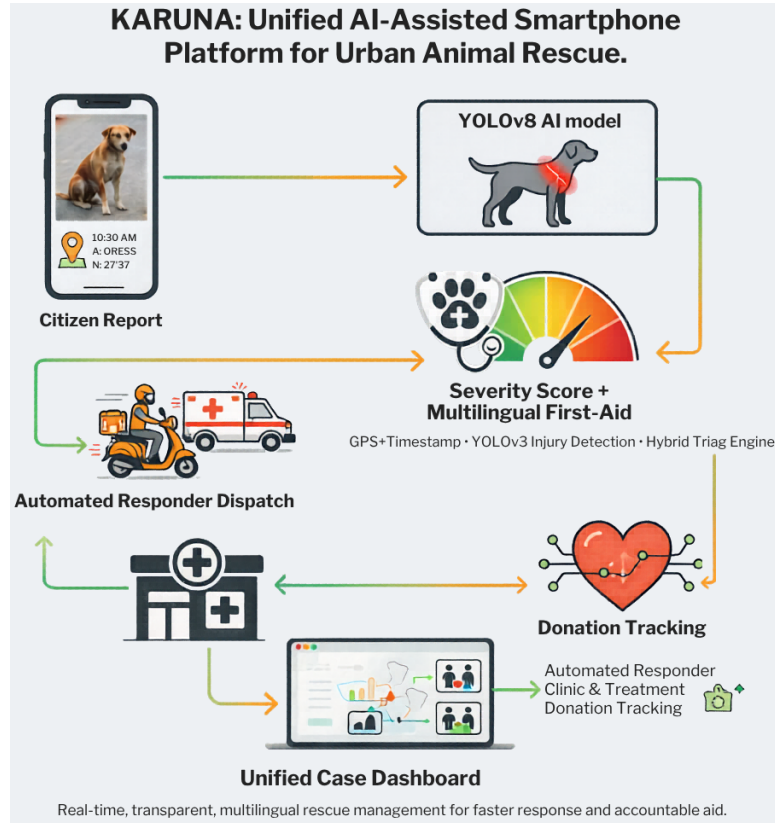


Figure 1: Overview of the proposed system.

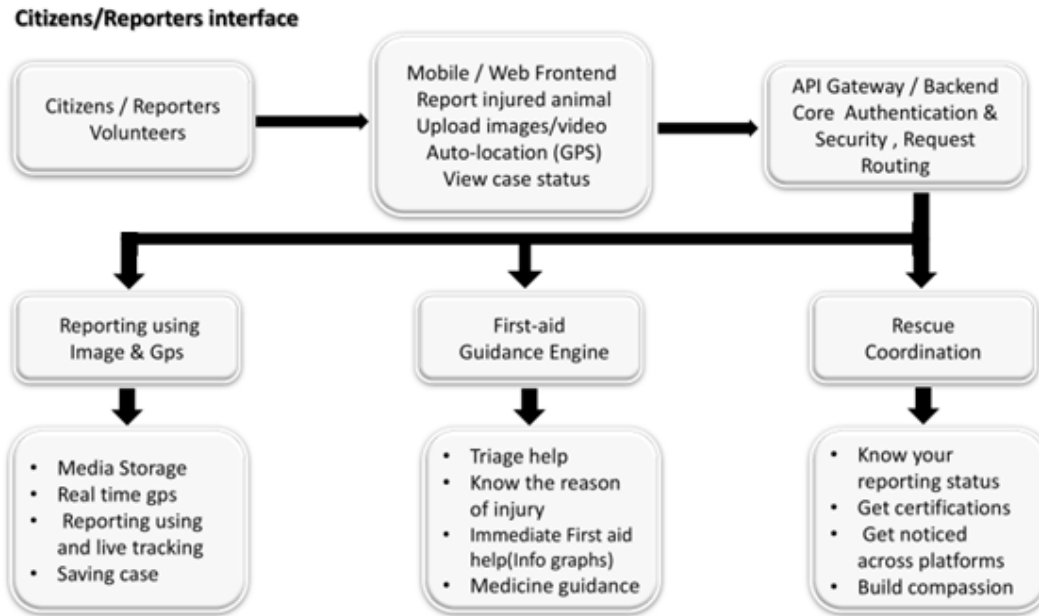


Figure 2: KARUNA Citizens/Reporters interface: from citizen report through mobile/web frontend and API gateway, to reporting, first aid guidance engine, and rescue coordination modules..

## 4.1 How the Platform Works

When a citizen photographs an injured animal, the app captures GPS and a timestamp automatically so the user does not need to do anything extra. That image goes to an AI model that identifies the species and any visible injury indicators, and comes back with a severity label: critical, urgent, or routine. Based on that, the app generates first aid steps in the user’s preferred language, so the case has something concrete to do while they wait. The report becomes a structured case record, and a matching algorithm finds the best available responder by weighing distance, current caseload, and relevant skills, then sends a notification.

The responder accepts, gets a map link, and starts logging updates as the case moves forward collected, at clinic, under treatment, discharged. All of this is visible to the original reporter in real time; they are not left wondering whether anyone came. For high cost treatments, a casespecific donation page goes live automatically. Every rupee that comes in is logged against that animal’s actual bills, not a general fund. When the animal recovers, it can be listed for adoption on the same platform, with adopter verification and postplacement checkins built in. All of it feeds into dashboards that municipal ABC programmes can use, including hotspot maps, response time trends, disease cluster signals, and real-time animal health data.

## 4.2 Key Components and Technologies

The citizen app is built in React Native or Flutter, which means one codebase covers both Android and iOS. Camera, GPS, offline queuing, multilingual first aid instructions these are the basics. The organisational side lives in React.js dashboards, each scoped tightly to the user’s role: dispatchers see cases and routing, vets see referrals and treatment records, admins see analytics and partner management. The backend runs as FastAPI or Node.js microservices with REST and GraphQL endpoints. Gemini handles the image identification tasks that are too heavy for an ondevice model. Auth is JWTbased, and rate limiting plus structured logging sit at the API gateway.

For data, PostgreSQL handles all relational records cases, users, treatment logs, donation transactions, adoption history. Firebase Firestore keeps the live case state for realtime dashboard updates. Media goes to AWS S3 or Google Cloud Storage, served via CDN. Routing is Google Maps with OSRM as a costconscious fallback. Notifications go through Firebase Cloud Messaging, with Twilio for SMS and WhatsApp when push is unavailable.

The AI stack is where the technical investment is concentrated. The injury and species detection model is YOLOv8, finetuned on photographs of Indian street animals covering the injury types that actually show up in urban rescue: open wounds, suspected fractures, active bleeding, emaciation, and eye injuries. The triage engine takes those outputs and runs them through a hybrid of veterinary decision rules and a gradientboosted ML classifier, producing a severity level and a ranked first aid sequence. The matching engine then uses that severity level alongside GPS coordinates to score

candidate responders. Payments go through Razorpay, PayU, and Stripe. Every donation writes an immutable event to a percase financial log, giving the transparency that blockchain-based systems have proven effective for charitable accountability without the complexity of an actual blockchain [12] [11].

### 4.3 Why the Idea Is Unique

It is not that any single component here is unprecedented. YOLOv8 exists. Donation ledgers exist. Rescue reporting apps exist. Platforms like Let It Wag [3], PetFinder [5], and Strayz [4] handle intake and routing; HSIApps [6] digitises field data capture for ABC programmes; systems modelling work [7] quantifies population management strategies. What none of these platforms do is connect image-based injury detection, automated severity-driven dispatch, treatment tracking, and per-case donation accounting into a single workflow for one animal. KARUNA makes the severity score from the AI triage engine a direct input into the matching algorithm. That means the urgency of the triage decision shapes who gets dispatched and how fast, without a human dispatcher making that call in between. Based on the reviewed literature, this direct coupling between automated image-based triage output and dispatch scoring has not been reported in any deployed urban animal rescue platform. The per-case financial ledger is similarly specific: donors can see the animal, the bills, and the outcome, not just a receipt confirming that money left their account. And because the platform is designed for multi-organisation, multi-city use from the start, its data accumulates in a standardised schema that is actually useful for the kind of epidemiological and welfare research that in [14] [15], researchers identify as urgently needed.

## 5 Technical Specifications

### 5.1 System Architecture

KARUNA runs as independently deployable microservices across four layers: client apps, API services, AI processing, and data storage. Each layer scales on its own, so a spike in media uploads during a flood related rescue surge does not drag down the case management API. Everything is containerised in Docker and deployed on AWS or Google Cloud with autoscaling on.

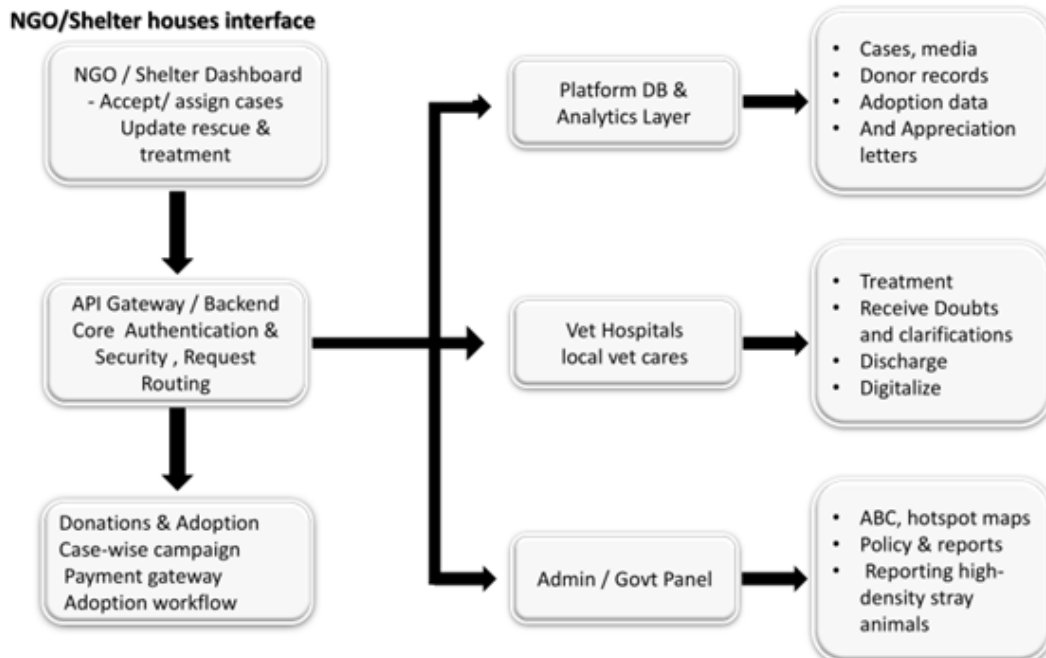


Figure 3: KARUNA NGO/Shelter houses interface: from NGO dashboard through API gateway to the platform database and analytics layer, vet hospitals, and admin/government panel modules.

### 5.2 Client Applications

The citizen app is React Native or Flutter one codebase, Android and iOS both covered. It asks for camera and GPS at install, keeps a local offline queue for poor connectivity areas, and preloads first aid content so it works even without data.

The organisational dashboards are React.js SPAs. Each role gets a different view: dispatchers see the live case map and assignment queue, vets see referrals and treatment records, and admins see platform analytics and partner management. Nobody sees more than they need.

### 5.3 Backend and APIs

The API layer is FastAPI or Node.js, with both REST and GraphQL endpoints depending on what each client needs. Gemini handles image identification calls that are too heavy for the on-device model. All traffic is HTTPS, with rate limiting, input sanitisation, and request logging enforced at the gateway. Auth runs on JWT with role-based access control across five user types: citizen, volunteer, NGO staff, veterinarian, and administrator.

### 5.4 Databases and Storage

PostgreSQL holds everything that needs to be auditable user accounts, case records, triage outputs, treatment logs, donation transactions, adoption records. Firebase Firestore handles the live state of active cases, pushing status updates to dashboards in real time without polling. Images and video go to AWS S3 or Google Cloud Storage and are served via CDN, which matters a lot for response times when the platform spans multiple cities.

### 5.5 AI and Decision Engines

The detection model is YOLOv8, finetuned on a dataset of Indian street animal photographs tagged for the injury presentations that actually come up in urban rescue work: open wounds, suspected fractures, active bleeding, severe emaciation, and eye injuries. It outputs bounding boxes, a species label, and injury category tags. Those go into the triage engine, a hybrid of a veterinary decision tree and a gradient boosted classifier trained on historical case outcomes which produces a severity level and a ranked first aid sequence. The severity level then feeds directly into the matching engine, which scores candidate responders on distance, specialisation, and current opencase load. That direct connection between triage output and dispatch scoring is the design choice that makes this system different from everything else reviewed.

### 5.6 Maps, Routing, and Communication

Turnbyturn routing uses Google Maps Directions, with OSRM as a cost fallback at scale. Push notifications go through Firebase Cloud Messaging. For users without smartphones or stable data, the same alerts go via SMS and WhatsApp through Twilio. Receipts and case closure reports go by email through Amazon SES or SendGrid.

### 5.7 Payments and Transparency

Razorpay and PayU handle domestic payments, Stripe for international donors. Every completed transaction writes an immutable event to a percase financial log amount, timestamp, donor identifier, and the specific bill it offsets. The NGO and the donor can export this log as PDF or CSV at any time. While this append-only event log does not provide the cryptographic immutability of a blockchain, it delivers practical per-case transparency at substantially lower infrastructure complexity.

### 5.8 NonFunctional Specifications

Target p95 API response time is under 500 ms for standard operations. Triage calls, which go to an external model, should return within three to five seconds under normal load. Database and media storage scale independently of the API and AI layers, so a photo upload surge does not slow down case management.

## 6 Expected Outcome

The pilot runs in Vijayawada and two nearby urban centres chosen because the NGO network there is already active and municipal partners have indicated willingness to integrate with ABC planning data. Within year one, the targets are: median rescue response time down from roughly 24 hours to under two hours; treatment success rate at 85% or above; at

least 40% of recovered animals placed in verified adoptions; and donor retention defined as a second contribution within twelve months at 70% or above. The last target is the one most directly tied to the percase transparency model. Indian donors do not keep giving to generic NGO accounts. They give when they can see what happened.

## 7 Novelty

When someone finds an injured animal, no existing app tells them what to actually do with it. Current platforms use the photograph to route the report. KARUNA uses it to infer the injury type, classify severity, and generate specific first aid steps in the user’s language before anyone professional has arrived. That alone is new. But the more significant novelty is what happens next. The severity score does not just get displayed. It becomes an input variable in the matching engine’s scoring function. Urgency drives dispatch, automatically. No human dispatcher sits between the AI assessment and the field response. Based on the reviewed literature including deployed platforms such as Let It Wag [3], PetFinder [5], Strayz [4], and HSIApps [6], as well as academic systems reported in [1], and [2], no urban animal rescue system has reported this direct coupling between image-based triage and automated dispatch scoring.

## 8 Technology Readiness Level (TRL)

Table 6: Technology Readiness Level (TRL).

| TRL Range | Stage                   |
|-----------|-------------------------|
| 1–3       | Idea / Proof of Concept |

## 9 Target Beneficiaries

The animals come first. Everything else in this document is in service of the fact that a faster, better coordinated response means fewer animals die waiting. NGOs and rescue groups benefit too; they stop receiving the same case through five channels and start seeing a clean intake queue with cases already triaged and ranked. Vets receive referrals that arrive with images, location history, and a first aid log already attached; they can prepare before the animal walks through the door rather than starting from zero. Citizen volunteers get something they currently lack entirely: confirmation. Right now, if you report a case, you have no idea whether anyone came. KARUNA closes that loop.

Municipal ABC units get real incident data, not estimates. That changes how sterilisation and vaccination routes get planned. Donors get per-case receipts with outcomes, charitable accountability shows is meaningfully better for retention than a generic thankyou email. And researchers, epidemiologists, One Health practitioners, urban ecologists get a structured, geographically tagged, continuously updated dataset that does not currently exist anywhere for Indian urban street animals.

## 10 Legal Framework

A few legal frameworks are directly relevant and worth being precise about. On animal welfare, the Prevention of Cruelty to Animals Act, 1960 sets the care baseline that all rescue activity on the platform must meet. The Animal Birth Control (Dogs) Rules, 2023 are also relevant: they place responsibility for stray dog management on local bodies and explicitly prohibit relocation. KARUNA’s ABC integration is designed to feed into those programmes, not create a parallel structure.

On data, the Digital Personal Data Protection Act, 2023 applies to everything the platform collects location records, photographs, contact details of reporters and volunteers. The design principle is minimal collection: location is retained only for the duration of an active case, and all research exports strip identifying information. On fundraising, NGOs collecting donations through the platform need to be compliant with FCRA for foreign contributions and with their Income Tax Act obligations as charitable organisations. KARUNA itself does not hold funds; money settles directly to the NGO’s registered account through the payment gateway, with the percase log as the audit trail. On public health, the data sharing features for municipal ABC units operate within state animal husbandry and ULB guidelines for stray animal management and zoonotic disease reporting not outside them.

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